

White Paper

**Implementing Multimedia Messaging
Within A Comprehensive Messaging Strategy**

Table of contents

1	Preface	2
2	Mobile Messaging	3
3	SMS: A Revenue Generator	4
3.1	Youthful generators	4
3.2	Premium services	5
3.3	Attracting prepaid	6
3.4	... and roamers	6
3.5	Fair weather	6
4	MMS: The Next Goldmine	7
4.1	More fun and games	7
4.2	Enhanced Messaging Service: an intermediate step	7
4.3	Multimedia Messaging Service: the ultimate step	8
4.4	Emerging user model	8
4.5	Pleasure with business	9
4.6	Pricing model	10
4.7	Money spinners	10
5	A Comprehensive Messaging Strategy	12
5.1	Common approach	12
5.2	Total messaging chain	12
5.3	Tactical weapons	13
5.3.1	Effective service: increased revenue	13
5.3.2	Efficient service: increased profit margins	13
5.4	Critical operational and IT issues	14
5.5	Implementing MMS	14
5.5.1	Power centre	15
5.5.2	Interconnect and roaming	15
5.5.3	Realtime secure prepaid billing and rating system	15
5.5.4	Seamless connectivity	16
5.5.5	Customer service and marketing tools	16
5.5.6	Content and killer apps	17
5.5.7	Reusing components	17
6	An MMS Implementation Scenario	18
6.1	An MMS solution	18
6.1.1	Speedy scalable MMS Centre	18
6.1.2	Realtime prepaid billing and rating	19
6.1.3	Connectivity	20
6.1.4	Personalised customer service	20
6.1.5	Content and Premium MMS applications	20
6.1.6	Integration, connectivity and sharing	21
6.2	Meeting the criteria	21
7	Outlook For MMS	22

1 Preface

As growth in mobile subscriptions begins to flatten, operators need to differentiate themselves to reduce churn, attract new subscribers and find new ways of stimulating their average revenue per user (ARPU).

CMG Wireless Data Solutions, a leader in mobile messaging solutions, is firmly convinced that mobile data, and in particular mobile messaging, provides an answer to the financial challenges faced by operators today. Moreover, CMG strongly believes that mobile messaging should be deployed in a broad and comprehensive way, encompassing current and future mobile messaging services, as well as using a set of uniform and consistent guidelines. Empowered by the twin drivers of effectiveness and efficiency, this strategy is based on quality of service and the user experience, as well as minimising costs to generate higher revenues and broader profit margins for operators and service providers.

This white paper, directed at mobile operators and service providers, describes the current situation of mobile messaging; investigates MMS; defines CMG's vision and strategy on implementing MMS; and investigates an implementation scenario.

2 Mobile Messaging

Operators and service providers in search of alternative growth strategies are increasingly turning to mobile data services. And there seems to be a ready-made market for them: in the business sector, travelling workers suffer the frustration of being cut off from their corporate information systems; and consumers, especially in Europe and Asia (where the mobile-device penetration is higher than that of personal computers), have grown accustomed to using their mobile devices for basic communication.

Mobile data services are about highly personalised services on small mobile devices. Messaging, the most personal of all data services, is therefore generally at the heart of mobile data services. The success of the Short Message Service (SMS), the dominant mobile messaging service, has generated industry wide interest in this application area.

As new technologies arrive on the scene, so new messaging services will start appearing. The first, Enhanced Messaging Service (EMS), will enhance the text message with basic multimedia content. This will be followed by the Multimedia Messaging Service (MMS), offering ultimately exciting functionality and full-multimedia content to subscribers, and an array of business opportunities to operators and service providers. Of course, the introduction of a new service does not mean the imminent end of existing ones; they will continue to be deployed in parallel and enjoy popularity. It does mean, however, that new IT infrastructure will need to co-exist and communicate with existing infrastructure and operators will need an implementation strategy that drives the whole messaging operation.

Let us first review SMS and the factors that have made it so successful.

3 SMS: A Revenue Generator

3.1 Youthful generators

Driven mainly by youth as a part of its lifestyle, the Short Message Service, which allows mobile subscribers to send and receive short text-messages of up to 160 characters, remains extremely popular and continues to break records. At Telkomsel, a prominent Indonesian operator, SMS traffic grew ten-fold after it upgraded its SMS infrastructure.

SMS has proven itself as a highly successful well-established concept: the technology and business model have been tried and tested. However, the issue is: how to maximise revenues from investments already made in SMS.

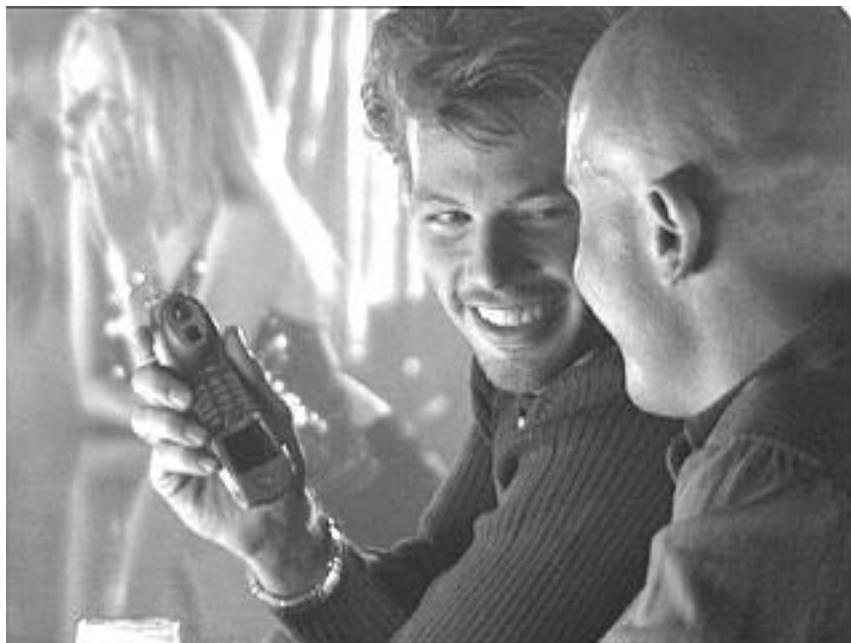


Figure 3.1: SMS is particularly popular in the 15–25-years age segment

SMS continues to increase the average revenue per user (ARPU). A mobile communications newsletter¹ recently revealed: “SMS is still far and away the key driver behind [mobile] data revenues.” Its top mobile-data revenue earners are Smart Communications (at 28 percent of ARPU) and Globe Telecom (at 18 percent), both of the Philippines, handling daily average-volumes of 90 and 72 million messages, respectively. The next-highest ranking operators are Telenor (also 18 percent) of Norway and D2 Vodafone (at 16.2 percent) of Germany. In these cases, most of the revenue is from SMS. Significantly, NTT DoCoMo is only ranked fifth place with 14.4

¹ 3G Mobile, July 25th 2001.

percent, most of which is generated by its famous mobile Internet data service, i-mode.

3.2 Premium services

Interpersonal messaging continues to contribute significantly—with revenues of US\$ 0.15 per message on average—to the bottom line of operators. Even markets, which early adopters have only just scratched the surface, have a long way to go. And there are novel ways—value-added SMS services, commonly referred to as Premium Short Message Services, or Premium SMS for short—to stimulate SMS usage for the operators and attract premium rates for service providers. Games, like the on-line, interactive, SMS-based variety announced by Scandinavian operator Telia and content-provider Moby earlier this year, for example. And chat groups. Yet another stimulus is location-based services: combining SMS messaging with user positioning (inherent in mobile telephony) to offer ‘nearest to’ services, like finding the nearest 5-star restaurant on the outskirts of a city or the nearest service station on the motorway.

And innovative m-commerce services. Such services allow subscribers to have their purchases from participating stores billed to their telephone account by sending their mobile phone company an SMS message. The phone company then processes the payment and sends a message back to the respective store confirming payment, thus completing the transaction. Key to the success of these services is near-instant response times and realtime rating and clearing-house mediation.

Then there is the German broadcaster who teamed up with three mobile operators and an entertainment company to offer ‘SMS TV’, a form of interactive where viewers watching a television program (‘Jede Sekunde Zählt’) use SMS to respond (several times until the right answer is declared) to a quiz question. (See Figure 3.2 for the results of the first ‘live to air’ program). For these interactive services, latency is totally unacceptable.

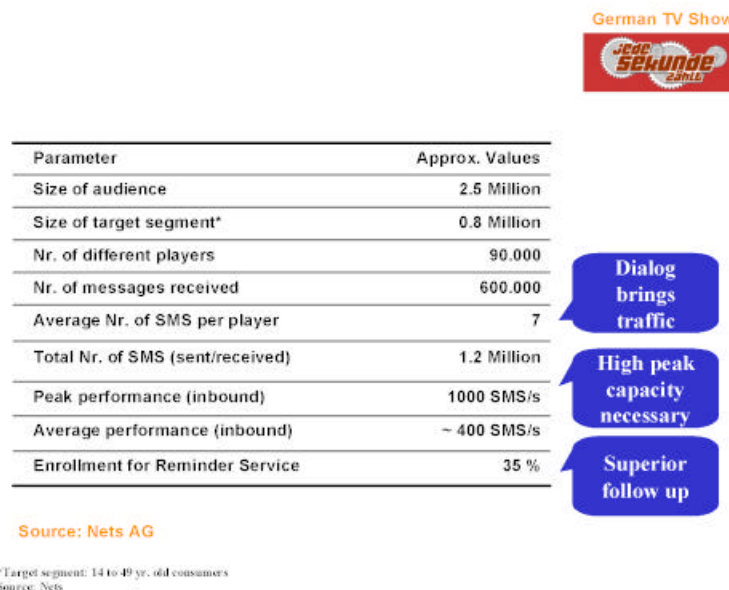


Figure 3.2: The effects of SMS TV in Germany

3.3 Attracting prepaid...

A significant way of increasing SMS messaging volumes is by introducing prepaid SMS. Young people, who form a large proportion of prepaid customers, are attracted to SMS. Furthermore, cost-conscious customers—again, mostly prepaid, youthful subscribers—tend to use SMS, which can be less expensive than making an equivalent voice call. Realtime billing is a prerequisite for offering prepaid SMS, because it is the only effective way of blocking the sending of further SMS messages once the subscriber's prepaid credit has run out. A significant increase in SMS traffic of 100 percent² (sometimes more) is not unusual when SMS for prepaid is introduced. Celcom, Malaysia's largest mobile operator is a case in point. It recently reported an unprecedented 800% increase in message volume within seven days after upgrading its infrastructure to handle prepaid SMS and SMS roaming.

3.4 ... and roamers

Offering SMS roaming and interconnect (allowing subscribers to exchange SMS messages with subscribers on other networks; and to send SMS messages when they are outside their home network) further stimulates SMS usage. In a report published last year, Mobile Lifestreams stated that adding national SMS-interconnect could lead to an increase of 50 percent in SMS message volumes.

3.5 Fair weather

And the outlook is good. The GSM Association's last updated SMS statistics reported a worldwide monthly volume of 20 billion messages to the end of June 2001. Then there is the Yankee Group's report on the Chinese mobile data sector. It stated in the highlights reported on August 2, 2001 that SMS will still be the dominant service over the next five years in China, a huge mobile market with enormous potential, thereby accounting for 60% of total data revenues (or US\$3.4 billion) for operators. Outside the commercial arena, SMS will continue to play an important role as principal bearer in WAP-Push and GPRS session-origination services.

² According to the *Messaging Metrics* report published by Mobile Lifestreams, March 2001.

4 MMS: The Next Goldmine

4.1 More fun and games

Mobile communications is evolving swiftly (see Figure 4.1) from a mainly traditional voice-driven communications industry into a personal information-oriented one, rich in multimedia content. Multimedia Messaging Service (MMS), the first mass-market application for new high-capacity mobile data networks, is due for launch in 2002 and set to make a significant impact on our personal and working lives. More importantly, it will build on the successful SMS experience, extending the boundaries for fun, games and entertainment, so much favoured by the youth, and enhancing the user experience even more.

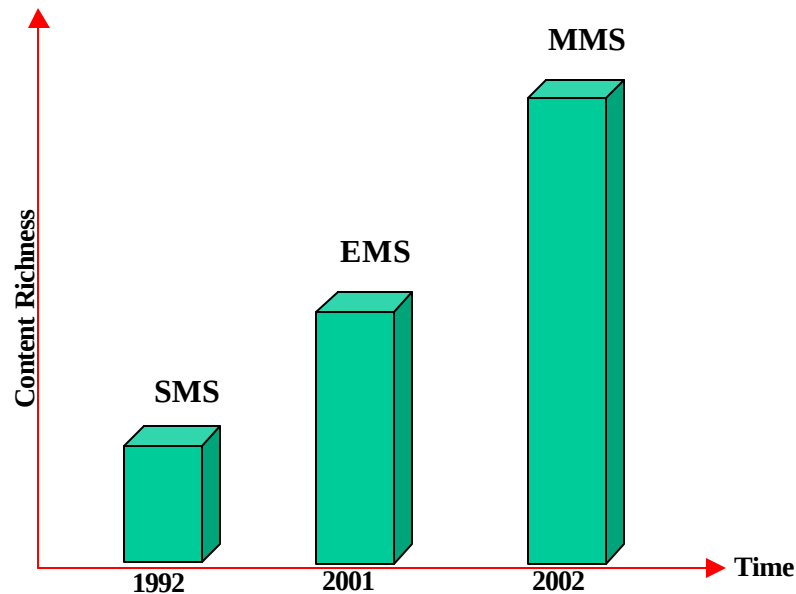


Figure 4.1: Adding value progressively to mobile messaging

4.2 Enhanced Messaging Service: an intermediate step

We saw the first move away from voice with the introduction of the text-based Short Message Service (SMS). The next step, Enhanced Messaging Service (EMS), has rejuvenated the popular SMS, 'enhancing' it with simple media features. Using EMS, mobile-phone users can liven up their text messages using several fonts, bitmap images, melodies and animations, as well as collecting and swapping logos and ring tones between the handsets of some of the world's leading suppliers.

There are signs this messaging service is now ready for take off. Major market-players, Alcatel, Ericsson, Motorola and Siemens, have all announced they will implement EMS and co-operate to ensure interoperability between their products and

in the evolution of the EMS standard. EMS can be seen as an extension of SMS with each of its messages being handled as a series of concatenated SMS messages. This means that operators can continue to use (and more importantly, derive income from) current 2G networks and SMS processing facilities. However, it is crucial that operators ensure their infrastructure and SMS centres have the necessary capacity to handle the increase in message traffic. With its attractive revenues and limited costs, EMS should be seriously considered by operators as an intermediate step before MMS arrives and takes off.

4.3 Multimedia Messaging Service: the ultimate step

As the mobile industry introduces advanced high-speed, data-centric, always-on telephone networks, like GPRS and 3G, and with global standards being set by the 3GPP (3G Partnership Program), the dream of stepping into full multimedia messaging becomes reality. MMS will enable the transmission of messages containing high-resolution colour pictures, full stereo audio sound, drawings, animations and (streaming) video—in any combination—between mobile devices and from applications to these devices.

4.4 Emerging user model

The mobile device has become 'personal' and ever-present; almost a part of our dress attire. Initially, mobile users just needed to communicate or be contactable at all times. Now, they need to inform and be kept informed. A new powerful and demanding mobile-user model is emerging, where instant communications and information are taken for granted. Messaging is a natural way to communicate: instant, location-independent, personal and fun.



Figure 4.2: First and foremost, Multimedia Messaging is about fun

Messaging can also be generational. Today's youth, grown up on video games and the Internet, has adopted SMS as an icon and is expected to embrace MMS as well. And there is potentially a new user group: subscribers who consider recording a voice memo, for example, easier than composing an SMS message. As business and work-related activities sometimes intrude in our private lives, mobile devices will be used increasingly for leisure and other personal activities, and the boundary between the traditional business and consumer user will start to blur. If the SMS experience is

used as a model, the uptake of MMS will most probably start among consumers, especially the young, and certain 'niche' professionals, before business users discover its attraction and switch over in increasing numbers.

4.5 Pleasure with business

MMS messages can be fun and exciting, like downloading celebrity video clips of famous pop artists, sharing holiday experiences by attaching a spontaneous snapshot with verbal annotations and sending it to family and friends. Then there is the question of convenience. Some subscribers will be delighted with the simplicity of an MMS voice-memo instead of typing in a text SMS message. And utility. Perhaps not absolutely necessary, but it will be very handy to be able to receive a map of the local area alongside verbal or textual instructions, for example. MMS is also for professionals. A journalist working under difficult conditions in the field could file a report with images and sound.

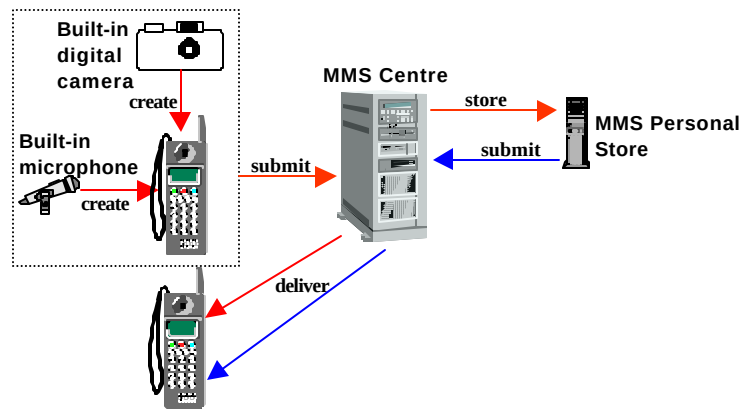


Figure 4.3: Some typical MMS applications

Driven by MMS, other services, such as news, sports and location-based information, are also expected to benefit from new functionality and capabilities. Subscribers can expect new applications, like Internet-based interactive video games, or having their simple text messages embellished with a celebrity jingle, animation or video clip. And subscribers will also have access to a Personal Store to save their favourite MMS messages, as well as, a photo album and library of images and sounds.

4.6 Pricing model

Present connect-time pricing models (designed with voice in mind) are not suitable for data traffic because they tend to reward slow networks with higher payments by the subscriber; thus hampering mobile data services. Thankfully, this is about to change with always-on, high-speed networks, where the model is most likely to be based on data volume.

However, the MMS pricing model should be similar to the SMS one, in order to accelerate the introduction and acceptance of the service. Successful pricing will be on a per-message basis and where the initiating party pays. In addition, premium rates can be applied to high-value content and, to a lesser extent, to richer media formats.

4.7 Money spinners

MMS exploits subscribers' experience and familiarity with SMS to help it over the acceptance threshold. And it is being assisted by other technologies. For example, the rapid growth of digital imaging and instant (digital) photography—pictures delivered instantly—is just the impetus needed to set the multimedia ball rolling. Based on the SMS experience, the majority of MMS messages will contain self-created information, in some form or another. In many cases, this will be personal text with a digitised photograph attached; or a personal note with some novel embellishment, which can be downloaded from a website (service providers, take note). Moreover, coupled with a mailing service and a distribution list, a holiday postcard can be sent to all relatives, friends and colleagues in one single action.

For operators, MMS offers competitive differentiation and can induce customer loyalty, as well as attract content providers and subscribers. All of this is expected to

stimulate service usage, where operators will profit from a significant growth in revenue from higher message volumes and larger messages.

Due to be commercially launched in 2002, MMS is expected to reach critical mass by 2004.

5 A Comprehensive Messaging Strategy

5.1 Common approach

Mobile messaging services enjoy strong similarities and synergies. Both in marketing and implementation, SMS, EMS and MMS frequently share the same (or variations of the same) marketing approach and success factors, as well as IT and operational requirements. This calls for an implementation strategy common to all three services. It consists of a 'generic' core (which it shares with all three services); and a 'specifics' component containing issues (mainly in such areas as content-type, complexity, dimensions and magnitude) peculiar to the service being implemented. The generic part of this strategy is largely based on lessons learned with SMS.

5.2 Total messaging chain

As new services come onboard, existing services, like SMS, will continue to enjoy popularity (refer to Figure 5.1) for sometime before tapering off. This means that new services should be introduced in parallel with current ones. Furthermore, messaging services and their associated IT infrastructures are interrelated and interdependent and new services should not be implemented in isolation of the total messaging operation; their IT infrastructures will need to co-exist and communicate with existing components within an operator's messaging operation. This calls for an integrated approach.

Service Type	Percentage of Total Mobile Messaging Market in the Year:					
	2000	2001	2002	2003	2004	2005
SMS	90	83	67	50	31	25
EMS	0	2	6	9	12	7
MMS	0	0	2	7	16	32
Others	10	15	25	34	41	36

Figure 5.1: How the mobile messaging market is expected to grow³

Crucially, operators need to look at any messaging service implementation within the wider context of a comprehensive mobile messaging operation (see Figure 5.2): not only investigating requirements for the service being implemented; but also taking into account all current (legacy) and future messaging services and relevant infrastructure. Issues such as integration, compatibility, and component reusability and sharing should therefore be investigated when considering a new service. A comprehensive strategy—one with a common and integrated approach—ensures uniformity, consistency and seamless access across all services, and also increases efficiency by, for example, sharing or reusing components or integrating operations.

³ Mobile Lifestreams' *Messaging Metrics* report, March 2001.

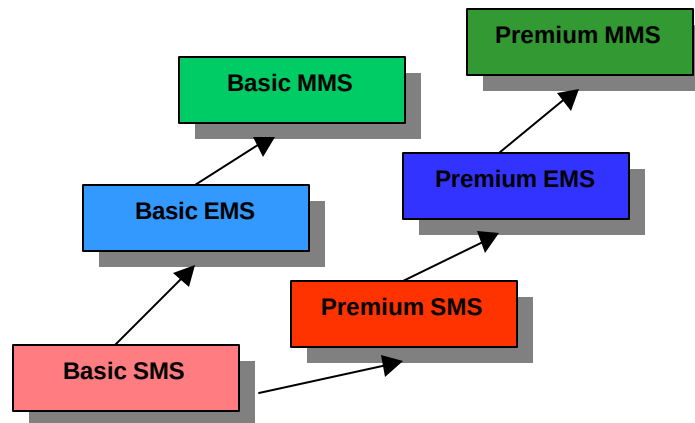


Figure 5.2: A growing messaging operation

5.3 Tactical weapons

What should drive this strategy? Rightly concerned with the bottom line, operators consider a successful mobile messaging business as one where revenues and profits can be sustained and increased, while costs are kept down. What is then key to a successful service and total messaging operation? Based on CMG's experience in the SMS market, the answer is effectiveness and efficiency that translates into Quality of Service (QoS). The reasoning is as follows. Effectiveness increases service usage and reduces churn, leading to higher revenues; and efficiency lowers costs and increases savings, leading to broader profit margins.

5.3.1 Effective service: increased revenue

Effectiveness reflects user experience and satisfaction, relating directly to QoS) and gauged by the following criteria:

1. Performance and 'smoothness' (especially in the case of applications sensitive to 'jerky' reception);
2. Availability and seamlessness (anytime, anyone and anywhere);
3. User friendliness and satisfaction;
4. Richness (in content and functionality);
5. Attentiveness to customer needs (engenders loyalty).

Referring again to Figure 5.2, we see subscribers' expectations and demands in terms of service experience increase as we step up the services ladder. Simply put, it means the IT infrastructure has to deliver the high-level of capacity and performance associated with richer and more advanced, business-critical services.

5.3.2 Efficient service: increased profit margins

Efficiency, the second part of this two-prong approach, targets costs, keeping them to a minimum. This is achieved by:

1. Minimising cost of ownership;
2. Minimising loss through fraud, especially prepaid;

3. Reusing (with minimum changes) IT components or sharing them with other messaging services.

5.4 Critical operational and IT issues

Translating these effectiveness and efficiency criteria into IT and operational requirements, we see that operators need to address the following issues:

1. Message centre: scalability, capacity, reliability and unified operations are key issues;
2. Roaming and interconnect: extending service to travelling and 'external' subscribers;
3. Real time billing and rating system: makes prepaid subscription and Premium Services possible; fast, near-instantaneous charging and rating, fraud reduction and secure transactions are central issues;
4. Connectivity and integration: enabling seamless and rapid service deployment;
5. Customer service and marketing tools: tailoring services to the customer's specific needs;
6. Content and value-added applications: this represents a large portion of the attraction for subscribers;
7. Solution reusability: selecting solutions, like billing, customer care and marketing tools, which can be also used by other departments.



Figure 5.3: The ease of use of Multimedia Messaging will attract users even beyond the traditional SMS user segments

5.5 Implementing MMS

Demanding technical specifications and operational requirements add to the challenge of finding an IT solution that satisfies MMS subscribers' high expectations

of a quality user experience. Applying the effectiveness and efficiency criteria (outlined in the previous section) to MMS produces the following requirements:

5.5.1 Power centre

The exciting user experience of bandwidth-hungry multimedia comes at a high cost to network and message-processing capacity and performance. This means high-capacity networks (GPRS or 3G) and highly reliable MMS centres.

While high-capacity networks are expected to alleviate network congestion (GPRS networks will practically deliver speeds between 24 and 40 kilobits per second, and 3G networks even higher speeds, high-performance MMS centres will be required to handle the exponential rise in both multimedia-message size and volumes. As more-sophisticated mobile devices become available, the MMS message size is expected to grow from 10-100KB in the early years, and even further as video and audio streaming come online.

There is also the question of the 'smoothness' of the service. While delays in Basic SMS (person-to-person) may be tolerated, for applications, such as Interactive SMS, delays are unacceptable because they drastically degrade the user experience. Considering that even Basic MMS could contain streamed video or audio content—applications especially sensitive to 'jerky' reception—latency becomes an even bigger issue with MMS.

Compare the current throughput requirement (in the order of megabits per second) for high-performance SMS centres, with the typical throughput MMS centres are likely to demand (in the order of tens or even hundreds of megabits per second), and the need for powerful message-processing engines becomes clear and obvious. Furthermore, as MMS popularity and profitability rise, so will the demand from operators, service providers and subscribers for an uninterrupted service. This will mean high-availability centres that are also scalable and can be upgraded onsite with minimum disruption.

5.5.2 Interconnect and roaming

Highly available MMS also implies a service available anywhere. This means that an operator's MMS centre and billing system should be able to interconnect with other national and international networks and MMS infrastructure; thus enabling subscribers to exchange MMS messages with other subscribers in any network, even when roaming. Thus, an anytime, anywhere quality service.

5.5.3 Realtime secure prepaid billing and rating system

Prepaid subscribers have been seen as a driving force behind SMS and are expected to play a key role in MMS as well. Offering prepaid MMS means operators need to install a prepaid billing system that can match the performance and reliability of their MMS centres, and is effectively integrated.

Differentiated charging (rating) is critical for MMS because of a wider selection of Premium Services and a much longer and broader content classification. Furthermore, with a higher (average) transaction-value, revenue risks will also increase accordingly. Operators are also burdened with an average monthly loss of US\$ 500,000 in prepaid fraud, and security procedures need to be in place (including

in the prepaid billing systems) before launching m-commerce applications. This means secure prepaid systems which do billing and rating in realtime. This is not only to prevent the creation of fraud windows, but also operational bottlenecks. Finally, prepaid billing systems will need to cater to MMS roaming, requiring full mobile-originator and terminator charging.



Figure 5.4: Also business users will benefit from the rich experience offered by Multimedia Messaging

5.5.4 Seamless connectivity

With the expected uptake of external and internal applications, and the continued use of legacy services, like SMS, operators need to ensure their MMS centres can be connected to and inter-work with other application servers or message centres. This will heighten the user experience with a seamless path from one service to the next and with uniform, consistent and user-friendly interfaces.

5.5.5 Customer service and marketing tools

As a market, such as MMS, gets more sophisticated, so do its consumers. A recent article⁴ in *The Economist* says it all: "On the face of it, companies seem keener than ever to sell you their stuff... But why do you so rarely feel that you are getting special service, or that the company knows the faintest thing about who you are or what you really want... The right approach to retaining customers starts with trying to understand more about them, and then work out what to do with the knowledge." In other words, effective personalised customer care.

Furthermore, operators and service providers need to proactively 'track' the market to be able to anticipate and adequately respond to subscribers' needs (tailored marketing): "The winning operators will be those with, for example, a thorough and

⁴ 'Keeping the customer satisfied', *The Economist*, July 14 2001.

accurate understanding of what services the market actually wants, and the ability to market these services effectively," says Ben Donnelly, Mobile Telecoms Analyst at Frost & Sullivan⁵.

To be able to offer prompt proactive (not only react to complaints) personalised customer service, operators need efficient and effective tools that track and analyse the market, as well as customer behaviour and preferences, as well as have access to relevant information (billing history, for example). In this case (as for realtime billing) 'effective' means the marketing tools must be able to keep up with the rate at which the MMSCs process messages.

5.5.6 Content and killer apps

Content is the visible face of MMS, whose success relies heavily on what content is available to subscribers and how it is presented and manipulated. In other words, if operators are to take MMS beyond interpersonal messaging (in itself a valuable service), they need to offer subscribers high-quality applications (Premium MMS), which use and manipulate multimedia content to its fullest, combining it with functions and features, such as location-sensitivity and m-commerce, to further enhance and exploit it.

5.5.7 Reusing components

Not peculiar to MMS but nevertheless important, many generic functional components, such as billing, customer care and marketing, could conceivably be reused, shared directly, or with minimal adaptations. This could significantly reduce the cost of ownership through, lower capital costs, but also sharing maintenance, support and operational expenses.

⁵ Frost & Sullivan's Profiles Of European Mobile Operators – Competitive Positioning And Strategies Leading To 3G Report, March 2001

6 An MMS Implementation Scenario

6.1 An MMS solution

Let us consider an example of how MMS can be implemented and review it in the context of the framework already discussed.

CMG Wireless Data Solutions has been involved in mobile messaging since the early 1990's when it was asked to design and develop a SMS Centre for a Nordic consortium. Since then, it has become a leading global supplier of wireless messaging, mobile Internet and customer care and billing solutions for the telecommunications industry, and is renowned for its quality of service solutions.

Building on other market successes, most notably in SMS, CMG offers a complete, turnkey MMS solution, which provides many benefits over multi-vendor solutions in the area of quality of service, cost of ownership and manageability. Furthermore, CMG maintains a policy and reputation for consistent quality—high scalability, capacity and reliability—across all solutions.

By opting for separate, dedicated and best-of-breed components, CMG offers a quality solution made up of quality parts. This contrasts with the 'Swiss army knife' approach available in the market today, which comes with full functionality, integrated and 'under one roof'. While this does not actually meet the specialised needs of the operator in implementing an MMS strategy, the 'Swiss army knife' solution might appear to be 'handy' and less time consuming to implement. However, its individual functions typically lack quality, flexibility and scalability.

6.1.1 Speedy scalable MMS Centre

At the core of the solution is its Multimedia Messaging Service Centre (MMSC), which processes MMS messages to and from an operator's network, as well as external networks and applications.

The MMSC's functionality is distributed over five main modules:

1. Http Receive: receives MMS messages from mobile devices connected to the operator's own network;
2. Http Send and Message Store: has a dual function. It sends MMS messages to mobile devices connected to the operator's own network; and temporarily buffers (stores and retrieves) all MMS messages for further handling;
3. SMTP Receive: receives MMS messages from external operators' networks or Internet applications;
4. SMTP Send: sends MMS messages to external operators' networks or Internet applications;

5. MMS User Database: maintains extensive subscriber information and service profiles containing subscriber preferences and options; also shared by such functions as prepaid billing, customer care and marketing.

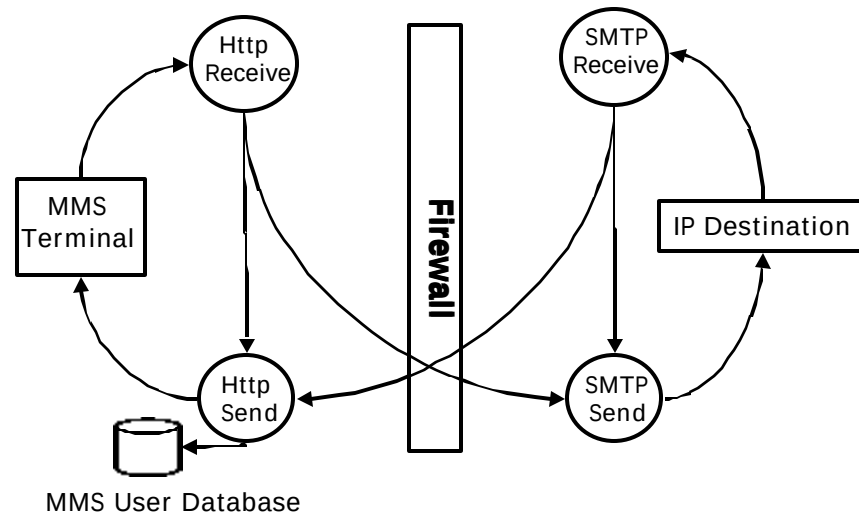


Figure 6.1: CMG's MMSC at work.

This distributed, modular architecture (see Figure 6.1) makes it seamlessly scalable to virtually unlimited performance requirements—only physical and network constraints, limit its size. In line with other CMG products, MMSC's performance and capacity are matched by its reliability (which translates into availability). It has no single point of failure; the system as a whole can continue to function in the event of network failures. There is redundancy in processors and critical devices, and each function module has the capability to operate independently from the other modules in the face of partial network and system failures. Furthermore, hardware components meet NEBS Level 3 standards for reliability and availability of 99.999%. This means any component failure will never lead to data loss. The low mean time to repair also means full service capabilities are quickly restored. Cost of ownership is low because running and maintaining the MMSC is virtually automated. Human intervention is only required in the event of hardware failure or new software installations. Even then, upgrades to the MMSC can be done online and while the system is still running.

CMG's MMSC comes with rich functionality. It supports a number of user features, like legacy-handset support, with message filters and SPAM and virus control.

6.1.2 Realtime prepaid billing and rating

Crucial to an MMS business, CMG's Payment Broker offers a flexible, high-performance (capable of 2000 transactions per second) prepaid rating and billing

solution. Its high-speed, realtime billing, rating (for Premium MMS differentiated charging) and credit checking prevents billing bottlenecks and eliminates prepaid fraud. In addition, the Payment Broker offers such advantages as MO and MT charging, full roaming support and secure m-commerce transactions (with encryption, payment authentication and digital signature), and has been designed to off-load vital network components.

6.1.3 Connectivity

For content and service providers, the MMSC offers a 'large account' or corporate interface, offering a secure MMS Internet-connection to an operator's subscriber base. In addition, the MMSC allows operators to act as service provider and offer their own (or trusted) applications and content services to their subscriber. Because these applications are secure sources, the connection between the MMSC and external application is fast, ensuring fast delivery of MMS messages and low latency, critical for interactive applications—and a quality user-experience. Control over content gives operators the flexibility of different charging schemes. Moreover, because operators offer these services directly, their profit margins are broader.

On the network side, CMG's Wireless Service Broker™ contains an excellent WAP Gateway solution, required in an MMS operation to connect mobile devices to the MMS Centre. The Wireless Service Broker™, designed according to CMG's scalability, reliability and high-performance specifications, can cope with growing data traffic levels and extensive features to create service packages for different end-user segments. In an MMS operation, its Pull and Push Service Brokers fulfil two key functions—the submission and delivery, respectively, of MMS messages—in the exchange of messages between the MMS Centre and MMS-enabled mobile devices.

6.1.4 Personalised customer service

Collectively called The Business Tools, this state-of-the-art software suite provides functionality to easily access and analyse raw operational and business data, transforming it into quality business knowledge. It is possible to store and analyse subscriber usage information to understand user trends and preferences and offer personalised and useful services, thus building customer confidence and reinforcing the relationship. Underpinned by the Customer Relationship Management concept,

The Business Tools provides an important means for operators to offer 'tailored and targeted' services, and hence increase profitability of the services as well as customer satisfaction. Timely traffic information means that operators can predict network peaks and bottlenecks more accurately. This means proactively managing the network, rather than reacting to problems after they occur, as is often the case. Furthermore, quality information means more accurate forecasting and better resource planning for operators, leading to more effective resource utilisation and lower costs.

6.1.5 Content and Premium MMS applications

In addition to providing operators with the tools to offer Premium MMS directly (using the MMSC Subscribed Services interface), CMG also offers off-the-shelf applications and content through its Partner Program. These applications have been integrated with CMG's MMS solution and can give operators a head start in marketing Premium MMS. And more importantly start making profits.

6.1.6 Integration, connectivity and sharing

Furthermore, CMG integrates its MMS offering with its other mobile messaging solutions, like SMS and Unified Messaging (UM). In addition, CMG connects its MMS environment to internal and third party applications and external infrastructures, and integrates its own products to legacy or third-party products. Not only does this make the whole mobile messaging operation seamless, but common functional components, like billing and customer care and marketing, can be shared by other services.

6.2 Meeting the criteria

CMG's MMS solution more than meets the QoS and efficiency criteria that are so key to operators implementing a mobile messaging strategy. Most notably, all critical components—such as MMSC, Prepaid Billing and WAP Gateway—are highly reliable and always provide enough capacity to handle peak message and transaction volumes, as well as the large messages MMS will generate. Not only does this solution prevent message bottlenecks; it also ensures high availability and a smooth user experience.

7 Outlook For MMS

High-capacity, data-centric networks, needed for MMS, are inevitable because of the expected drastic increase of (data) traffic and demand for more user-friendly and sophisticated services. Even though 3G networks are under attack and their commercial introduction could be delayed, GPRS could easily provide the necessary functionality and bandwidth in the meantime. This is confirmed in a recent issue of *The Economist*⁶ reporting on the GSM Association's 'M-Services Initiative' (backed by big operators and device manufacturers) to offer graphics and other multimedia on GPRS phones.

Moreover, MMS is the only mass-market application available shortly that fully utilises the added functionality, capacity and facility these networks have to offer. Not only does it provide operators with the justification and motivation to install such networks, MMS also offers a quick payback time. Moreover, it will not really require 3G networks until bandwidth-demanding features, like video, become available.

The conclusion is that MMS is on the horizon, confirmed by the UMTS Forum survey in April 2001 indicating that MMS subscriptions will reach 128.5 million by 2010 and revenues of US\$ 17.8 billion, worldwide. The incentives—a booming market and a potential cash cow—have already been forecast and the means—GPRS (and some 3G) networks and information technology—are already available.

To harvest these opportunities, CMG offers a pragmatic implementation strategy. It is defined in terms and objectives firmly rooted in the golden, business rule of the satisfied customer; based on simple arithmetic of economics; and aligned closely the financial ambitions of the operator (and service provider). Furthermore, CMG's strategy is based on its broad and highly successful experience in the mobile messaging market, and backed by a complete solution holding a pedigree and credentials, which stand up to the rigorous implementation requirements.

⁶ 'Nokia succumbs', *The Economist*, June 6th 2001